SCHOOL of EDUCATION and APPLIED PROFESSIONS

BUFFALO STATE • The State University of New York

COURSE SYLLABUS

Course Number: CIS451
Course Name: Data Structures
Semester: Spring 2026
CRN: 3365
Schedule (class time/day): MW
Mode of Instruction (Traditional, Hybrid, Hybrid Plus, Online): Traditional

Instructor: Dr. Gang (Gary) Hu
Email: hug@buffalostate.edu
Office Hours: MW 1-2:30pm
Office Location: 215 Tech Building
Phone: 716-878-5112

Course Description:
Fundamental concepts such as data organization, storage, and data search and retrieval concepts in software development. Explore various data structures, algorithms and corresponding methodologies of design, implementation, and analysis. Hands-on programming exercises provide students with skills in problem-solving techniques and algorithmic thinking.
Elective for junior/senior in CIS major

Prerequisite Courses: CIS 251 or CIS 361, for junior/senior of CIS major

Enrollment Restrictions: junior/senior in CIS major

Learning Outcomes for this Course: On completion of this course, students will be able to:

- SLO 1 Demonstrate time-space complexity of algorithms with the data structures.
- SLO 2 Create standard data structures, such as ArrayList, LinkedList, Queue, Stack
- SLO 3 Build programs using basic search and sorting algorithms
- SLO 4 Apply merge sort, heap sort, quick sort algorithms to solve practical problems
- SLO 5 Show the behavior of tree data structures
- SLO 6 Apply the recursive programming approach to applications
- SLO 7 Illustrate representations of graph data structures
- SLO 8 Explain the shortest path, breath first and depth first algorithms

Course Format

Course Requirements:
Required—n/a
Recommended Supplemental Materials—

1. Aditya Bhargava, Grokking Algorithms: An illustrated guide for programmers and other curious people, 1st Edition, Manning Publications, 2016, ISBN: 9781617292231.
2. Think C/C++, Allen B. Downey in PDF (Good intro book to C/C++ for the beginner programmer, also useful if Java was your first programming language)
   https://greenteapress.com/wp/think-c/
3. Problem Solving with Algorithms and Data Structures using Python
   https://runestone.academy/ns/books/published/pythonds3/index.html

Professional Materials and/or Fees for This Course—n/a
Assignments—4-6 individual home assignments, course projects.

Brightspace: Students have access through Brightspace for all courses in which they are enrolled.

Course Schedule: The following is a tentative course outline for this semester/term. The instructor may adjust content, assignments, and due dates to accommodate changing needs during the semester.

Week		Topic	Reading	Activity	Due
1	1/26	Course Intro	Ch 1		
		Data Structures as System Building Blocks			
2	2/2	Time vs. space trade-offs	Ch 2		
		Algorithm efficiency in practice (Big O overview)			
3	2/9	Arrays & Dynamic arrays			
		Memory locality and performance			
4	2/16	**President's Day: No classes**			A1 due
		Linked Lists			
5	2/23	Stack and queue patterns	Ch 3		
		Deques			
6	3/2	Recursion			A2 due
		Stack overflow			
7	3/9	Hash table			
		Caching and lookup tables			Topic due
8	3/16	**Summary**			
	3/20	Midterm Grade Due		**Exam 1**	
9	3/23~ 3/28	**Spring Recess - No Classes**			
10	3/30	Seminar 1		Presentation	A3 due
		Seminar 2		Presentation	
11	4/6	Trees for Structured Data	Ch 4		
		Traversals			
12	4/13	Balanced Trees			
		AVL / Red-Black trees			A4 due
13	4/20	Binary Search Trees & Indexing	Ch 5		
		Degeneration and data ordering			
14	4/27	Graphs for Relationships			
		BFS/DFS			
15	5/4	Graph Algorithm (Dijkstra's algorithm)			
		Summary			Project due
16	5/11	CEP		Final Exam	

Evaluation and Attendance

Grades:

Grading Weights:

Category	% Of Grade	Grade Items (Learning Activities)	Associated Learning Objectives/Outcomes
Written Assign-ments	30%	4 ~ 6 assignments	• SLO 1~ SLO 8
Presentation	15%	Survey Report, Oral presentation Peer evaluation	• Clearly deliver tech. concepts and methodologies
Exam 1	20%	Chapter 1 ~ Chapter 3	• SLO 1 ~ SLO 4
Exam 2	30%	Chapter 1~ Chapter 5	• SLO 1 ~ SLO 8
Participation	5%	Attendance, Question/Answer	• SLO 1~ SLO 8
TOTAL	**100%**		

Grading Scale:

Letter Grade	% Grade
A	93%-100%
A-	90%-92%
B+	87%-89%
B	83%-86%
B-	80%-82%
C+	77%-79%
C	73%-76%
C-	70%-72%
D+	66%-69%
D	60%-65%
E	<60

- *Students may also be granted Withdrawal (W), if they withdraw from the course, or an Incomplete (I) should their circumstances permit.*

Attendance:
Attendance is taken initially and checked against official roster sheets. It may be taken into consideration in determining border line grades. Lectures may contain information not provided by the text and material will not be reviewed for habitual absences. Persistent unexcused absences or a prolonged unexcused absence will result in a grade of E for the course.

Note: This syllabus is subject to change at the discretion of the instructor.